- Charts: visualizing data
- Getting started with BigQuery

Machine Learning

These are a few of the notebooks related to Machine Learning, including Google's online Machine Learning course. See the full course website for more.

- Intro to Pandas DataFrame
- Intro to RAPIDS cuDF to accelerate pandas
- Getting Started with cuML's accelerator mode

Using Accelerated Hardware

- Train a CNN to classify handwritten digits on the MNIST dataset using Flax NNX API
- Train a Vision Transformer (ViT) for image classification with JAX
- Text classification with a transformer language model using JAX

Featured examples

- Train a miniGPT language model with JAX AI Stack
- LoRA/QLoRA finetuning for LLM using Tunix
- Parameter-efficient fine-tuning of Gemma with LoRA and QLoRA
- Loading Hugging Face Transformers Checkpoints
- 8-bit Integer Quantization in Keras
- Float8 training and inference with a simple Transformer model
- Pretraining a Transformer from scratch with KerasHub
- Simple MNIST convnet
- Image classification from scratch using Keras 3
- Image Classification with KerasHub

```
import pandas as pd

# Sample DataFrame
property_data = pd.DataFrame({
    "Property_ID": [101, 102, 103, 104, 105],
    "Location": ["Chennai", "Bangalore", "Chennai", "Hyderabad", "Bangalore"],
    "Bedrooms": [3, 5, 4, 6, 2],
    "Area_sqft": [1500, 2500, 1800, 3200, 1400],
    "Listing_Price": [7500000, 12000000, 8500000, 15000000, 6800000]
})

# 1. Average listing price in each location
avg_price = property_data.groupby("Location")["Listing_Price"].mean()
print("Average Listing Price by Location:")
print(avg_price)

# 2. Number of properties with more than four bedrooms
count_bedrooms = property_data[property_data["Bedrooms"] > 4].shape[0]
print("\nNumber of Properties with More Than 4 Bedrooms:")
print(count_bedrooms)

# 3. Property with the largest area
largest_property = property_data.loc[property_data["Area_sqft"].idxmax()]
print("\nProperty with the Largest Area:")
print(largest_property)
```

```
Average Listing Price by Location:
Location
Bangalore     9400000.0
Chennai       8000000.0
Hyderabad    15000000.0
Name: Listing_Price, dtype: float64

Number of Properties with More Than 4 Bedrooms:
2

Property with the Largest Area:
Property_ID            104
```

```
Location         Hyderabad
Bedrooms                 6
Area_sqft             3200
Listing_Price     15000000
Name: 3, dtype: object
```